Eigenvalues & Eigenvectors

Geometric Algorithms

CAS CS 132

Practice Problem

Suppose A *is a* 234×300 *matrix. What is the smallest possible value for* $\dim(\mathrm{Nul}(A))$*? What is the largest possible value?*

What is the smallest possible value for $\mathrm{rank}(A)$*? What is the largest possible value?*

Answer

Objectives

» Motivate and introduce the fundamental notion of eigenvalues and eigenvectors

» Determine how to **verify** eigenvalues and eigenvectors

» Look at the **subspace** generated by eigenvectors

» Apply the study of eigenvectors to **dynamical linear systems**

Keyword

Eigenvalues

Eigenvectors

Null Space

Eigenspace

Linear Dynamical Systems

Closed-Form Solutions

Motivation

demo

How can matrices transform* vectors?

» rotations
» projections
» shearing
» reflection
» scaling/stretching
» ...

* square matrices

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Today's focus

All matrices do some combination of these things

* square matrices

What's special about scaling?

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We don't need a matrix, it's just $\mathbf{x} \mapsto c\mathbf{x}$

If $A\mathbf{v} = c\mathbf{v}$ then it's "easy to describe" what A does to $\mathbf{v}$

Eigenvectors (Informal)

eigenvalue

$$A\mathbf{v} = \lambda\mathbf{v}$$

eigenvector

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$$\underbrace{A\mathbf{v}}_{} = \overset{\text{eigenvalue}}{\lambda}\underbrace{\mathbf{v}}_{\text{eigenvector}}$$

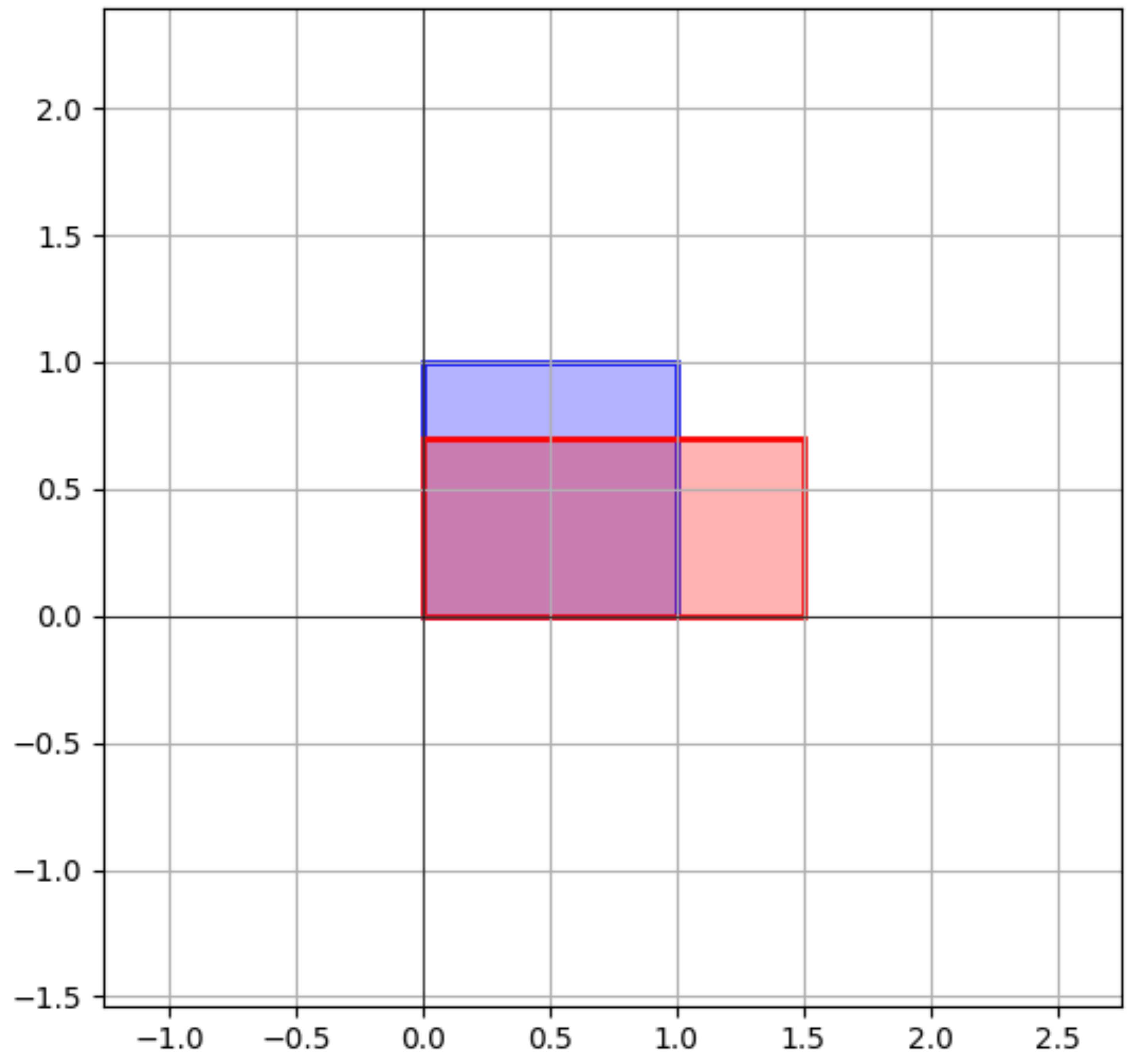
Eigenvectors of A are stretched by A without changing their direction

The amount they are stretched is called the **eigenvalue**

Example: Unequal Scaling

$$\begin{bmatrix} 1.5 & 0 \\ 0 & 0.7 \end{bmatrix}$$

It's "easy to describe" how unequal scaling transforms vectors. *It transforms each entry individually and then combines them*



Eigenbases (Informal)

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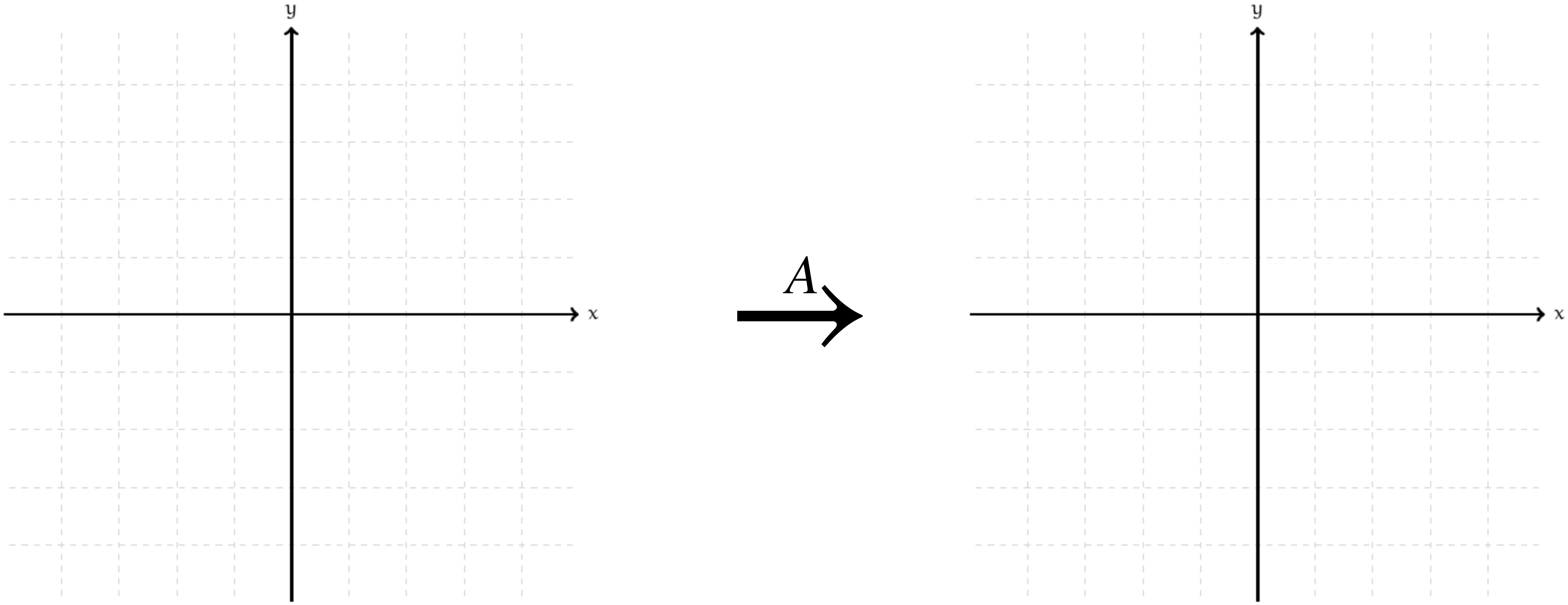
Suppose $\mathbf{v} = 2\mathbf{b}_1 - \mathbf{b}_2 - 5\mathbf{b}_3$ and $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3$ are *eigenvectors of* A

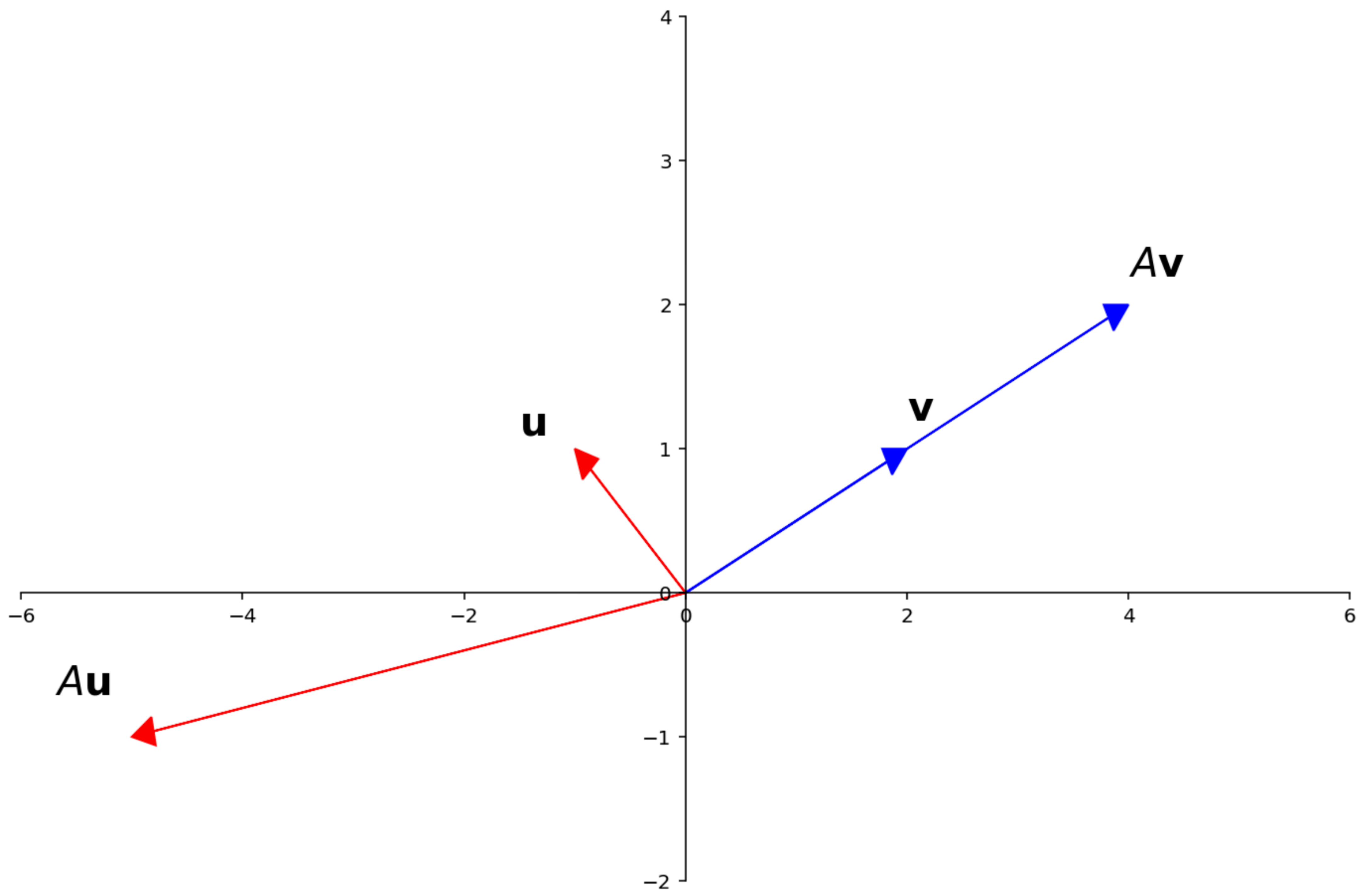
It's "easy to describe" how A transforms $\mathbf{v}$:

Eigenbases (Pictorially)



Eigenvalues and Eigenvectors

Formal Definition



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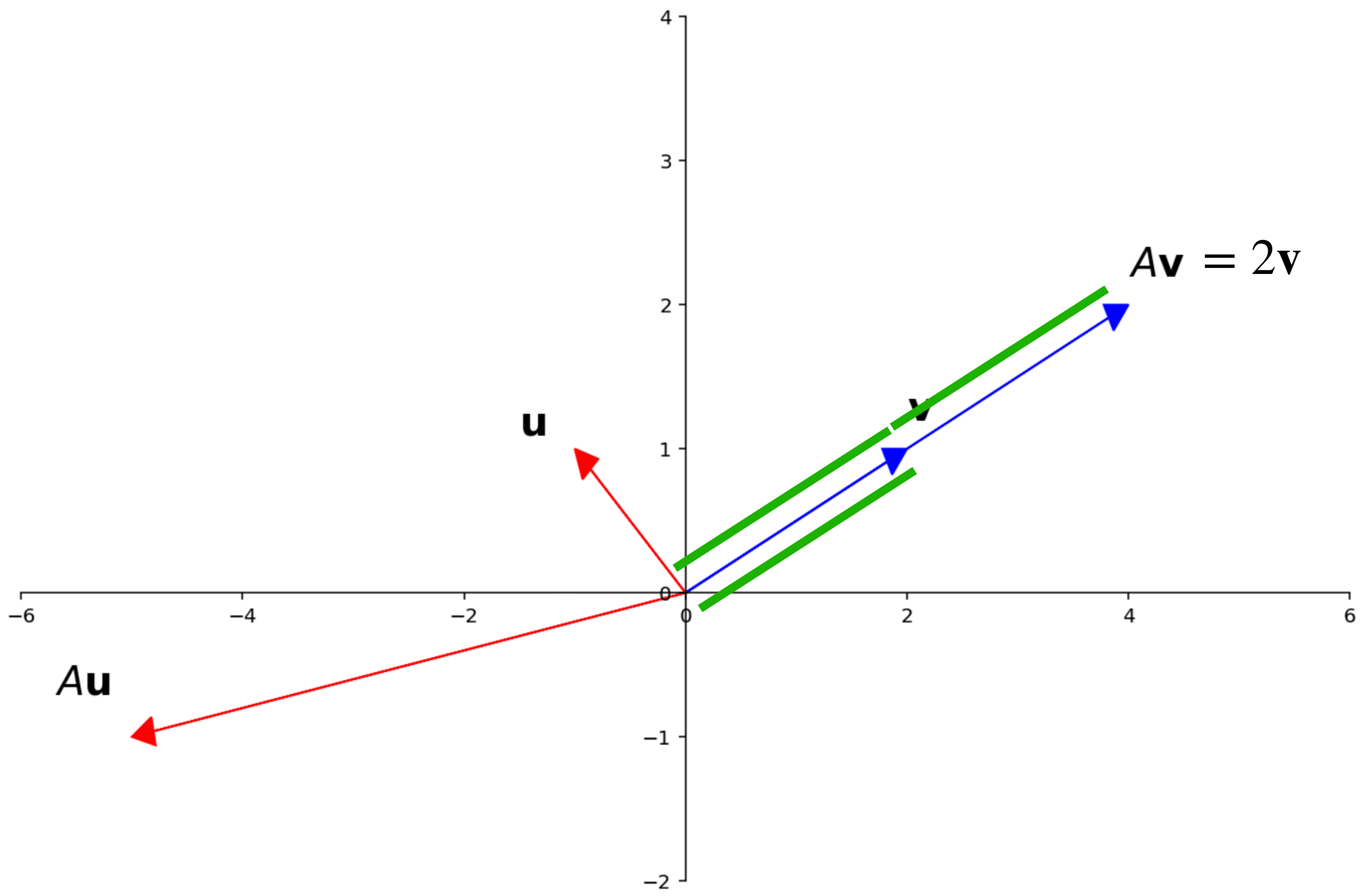

Def. A *nonzero* vector $\mathbf{v} \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$ are an **eigenvector and eigenvalue** for a $A \in \mathbb{R}^{n \times n}$ if $A\mathbf{v} = \lambda\mathbf{v}$

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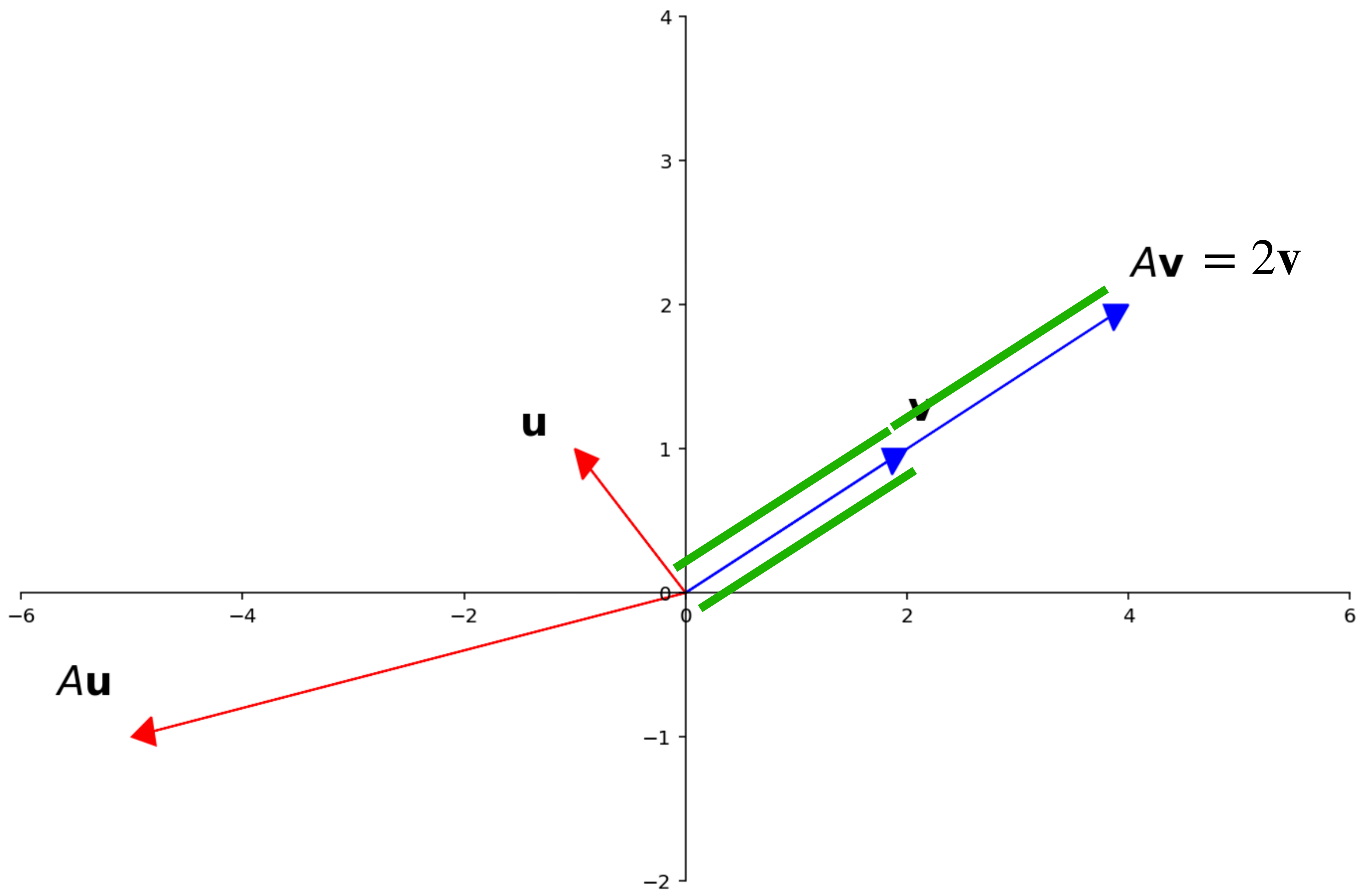

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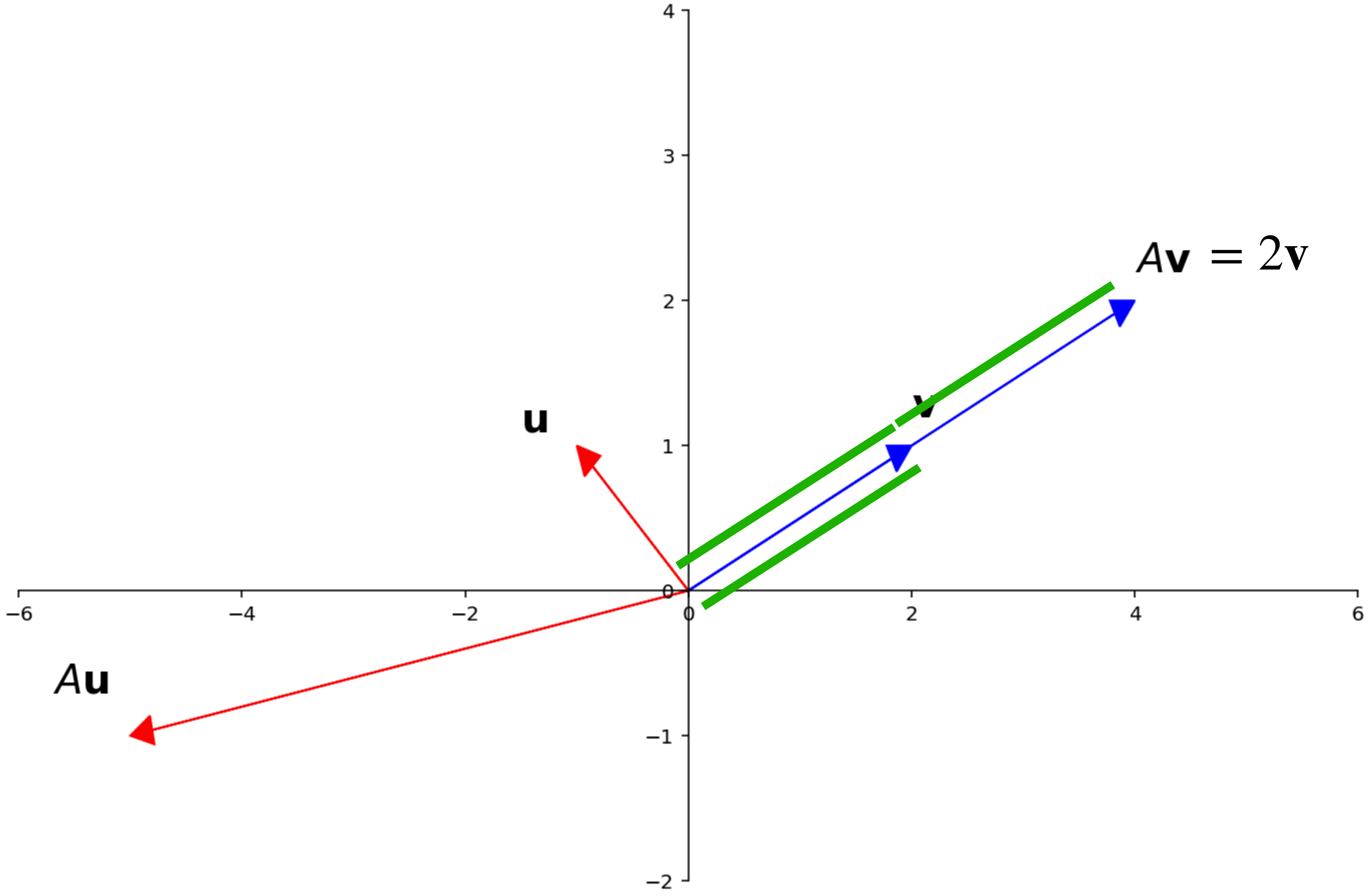

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Note. Eigenvectors *must* be nonzero, but 0 can be an eigenvalue

What if 0 is an eigenvalue?

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» $\mathbf{v} \in \mathsf{Nul}(A)$

» $\mathbf{v}$ is a nontrivial solution to $A\mathbf{v} = \mathbf{0}$

Extending the IMT (Again)

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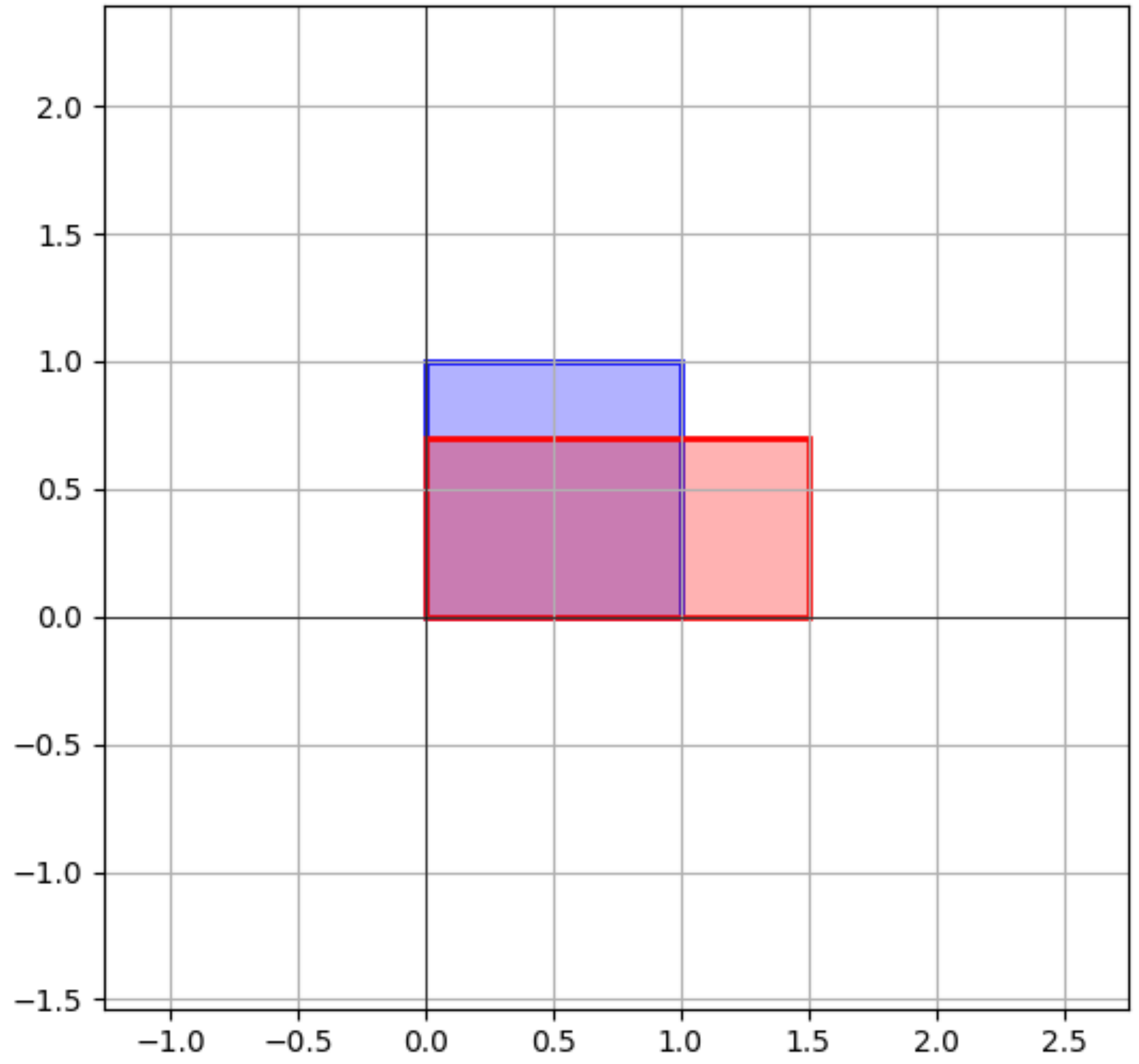
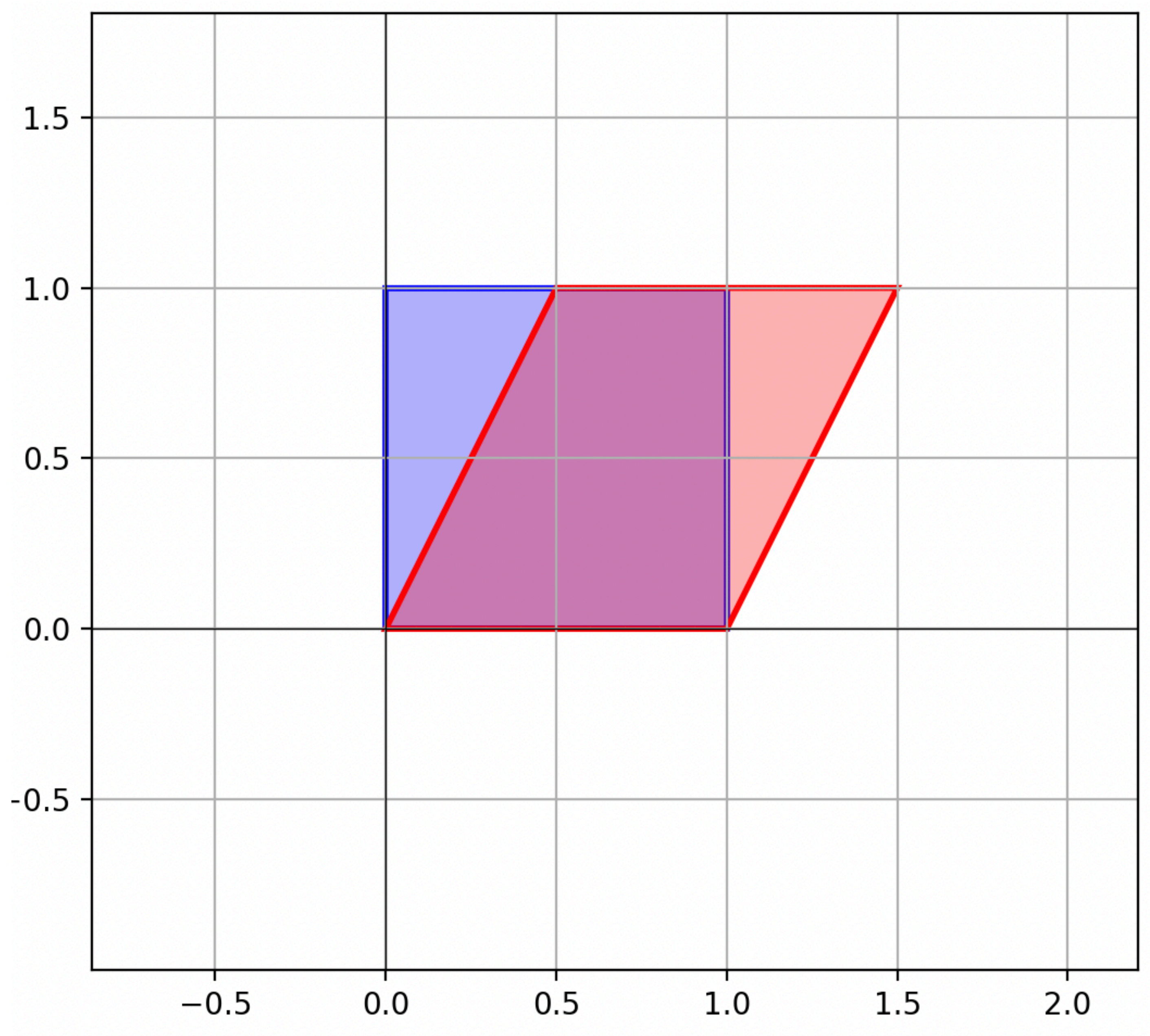


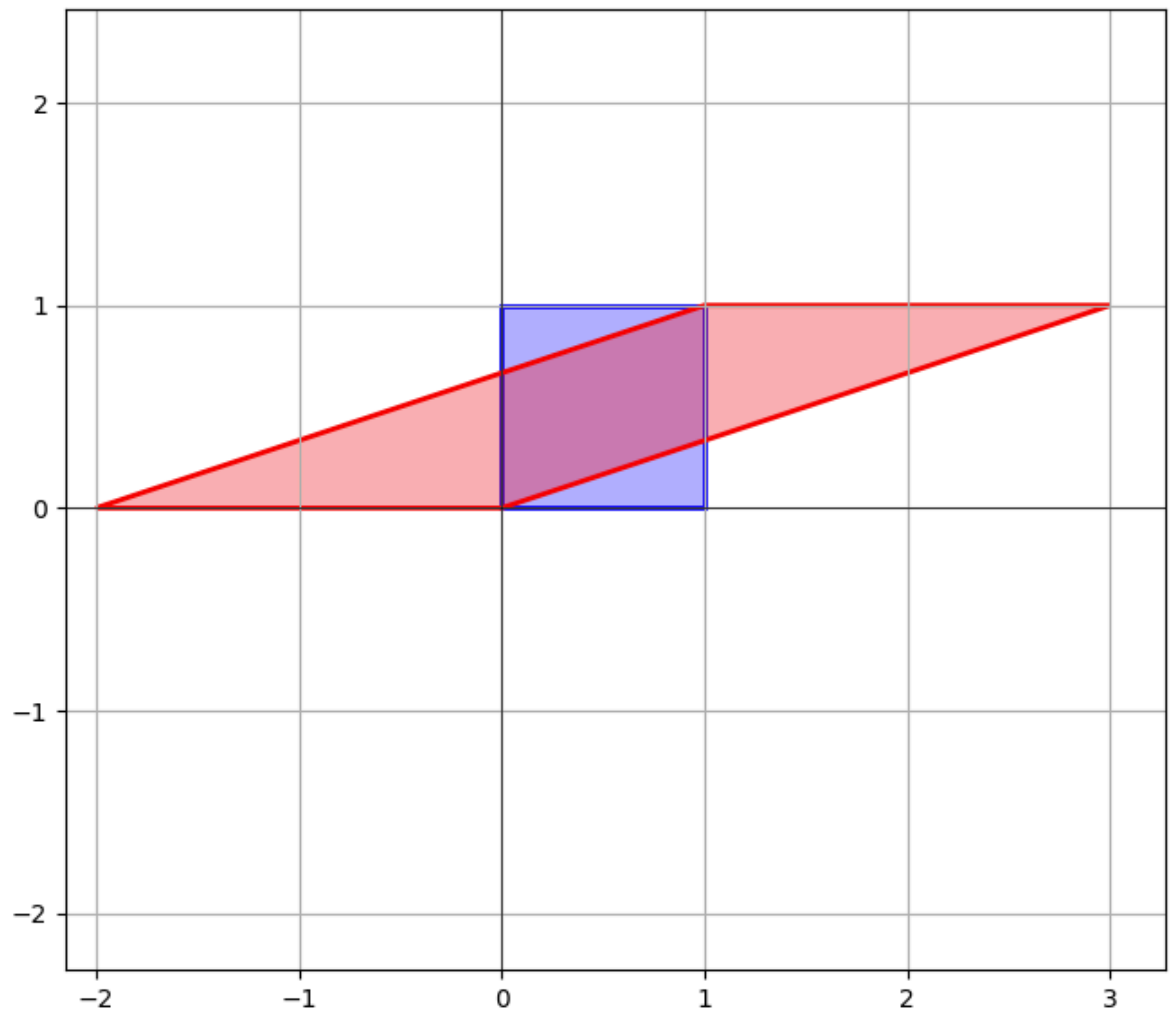
Example: Shearing

$$\begin{bmatrix} 1 & 0.5 \\ 0 & 1 \end{bmatrix}$$



Example

$$A = \begin{bmatrix} 3 & -2 \\ 1 & 0 \end{bmatrix} \quad \mathbf{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \mathbf{v} = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}$$



How do we verify eigenvalues and eigenvectors?

Verifying Eigenvectors

Verifying Eigenvectors

Q. *Determine if* $\begin{bmatrix} 6 \\ -5 \end{bmatrix}$ *or* $\begin{bmatrix} 3 \\ -2 \end{bmatrix}$ *are eigenvectors of* $\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$ *and determine their eigenvalues*

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A. Easy, work out the mv multiplication

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Before we go over how to do this...

Verifying Eigenvalues (Warm Up)

Q. Verify that 1 is an eigenvalue of

$\begin{bmatrix} 0.1 & 0.7 \\ 0.9 & 0.3 \end{bmatrix}$ (*Hint.* Recall Markov Chains)

A.

Steady-States and Eigenvectors

The Point: *Steady–state distributions are eigenvectors with eigenvalue* 1

$\mathbf{v}$ is a steady-state vector $\equiv$

$$\mathbf{v} \in \mathsf{Nul}(A - I)^*$$

$*$It must also be a probability vector

Verifying Eigenvalues

Q. Show that λ is an eigenvalue of A

A.

$$\mathbf{v} \texttt{ is an eigenvector for } \lambda \quad \equiv$$
$$\mathbf{v} \in \mathsf{Nul}(A - \lambda I)$$

Verifying Eigenvalues

Q. Show that 7 is an eigenvalue of $\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$

A.

Problem

Verify that 2 *is an eigenvalue of* $\begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix}$

Answer

$$\begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix}$$

How many eigenvectors can a matrix have?

Linear Independence of Eigenvectors

Thm. If $\mathbf{v}_1, \ldots, \mathbf{v}_k$ are eigenvectors for distinct eigenvalues, then they are linearly independent

So an $n \times n$ matrix can have at most n eigenvalues

Eigenspace

Fact. The set of eigenvectors for a eigenvalue λ of $A \in \mathbb{R}^{n\times n}$ form a subspace of $\mathbb{R}^n$

Eigenspace

Def. The set of eigenvectors for a eigenvalue λ of A is called the **eigenspace** of A for λ

It is the same as $\mathrm{Nul}(A - \lambda I)$

How To: Basis of an Eigenspace

Q. Find a basis for the eigenspace of A corresponding to λ

A. Find a basis for $\mathrm{Nul}(A - \lambda I)$

We know how to do this

Example

$$\begin{bmatrix} -2 & 0 & 3 \\ 1 & 1 & -1 \\ -4 & 0 & 5 \end{bmatrix}$$

Determine a basis for the eigenspace corresponding to the eigenvalue 1

How do we find eigenvalues?

How do we find eigenvalues?

We'll cover this next time...

Eigenvalues of Triangular Matrices

Thm. The eigenvalues of a triangular matrix are its entries along the diagonal

Example

$$\begin{bmatrix} 3 & 6 & -8 \\ 0 & 0 & 6 \\ 0 & 0 & 2 \end{bmatrix}$$

Determine the eigenvectors and values of the above matrix

Linear Dynamical Systems

Recall: Linear Dynamical Systems

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The **possible states** of the system are vectors in $\mathbb{R}^n$

Given an **initial state vector** $\mathbf{v}_0$, we can determine the **state vector** of the system after i time steps:

$$\mathbf{v}_i = A\mathbf{v}_{i-1}$$

(linear recurrence relation)

Recall: State Vectors

$$
\begin{aligned}
&\mathbf{v}_0 \\
&\mathbf{v}_1 = A\mathbf{v}_0 \\
&\mathbf{v}_2 = A\mathbf{v}_1 = A(A\mathbf{v}_0) \\
&\mathbf{v}_3 = A\mathbf{v}_2 = A(AA\mathbf{v}_0) \\
&\mathbf{v}_4 = A\mathbf{v}_3 = A(AAA\mathbf{v}_0) \\
&\mathbf{v}_5 = A\mathbf{v}_4 = A(AAAA\mathbf{v}_0) \\
&\quad\vdots
\end{aligned}
$$

The **state vector** $\mathbf{v}_k$ is the state of the system after k time steps

The Issue

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It's also difficult computationally because matrix multiplication is expensive

(Closed-Form) Solutions

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A **(closed–form) solution** of a linear recurrence relation $\mathbf{v}_{i+1} = A\mathbf{v}_i$ is an expression for $\mathbf{v}_k$ that does *not* contain A^k or previously defined terms

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A **(closed-form) solution** of a linear recurrence relation $\mathbf{v}_{i+1} = A\mathbf{v}_i$ is an expression for $\mathbf{v}_k$ that does *not* contain A^k or previously defined terms

In other word, it doesn't depend on A^k and isn't **recursive**

Example

$$\mathbf{v}_k = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \mathbf{v}_{k-1} \qquad \mathbf{v}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Solutions with Eigenvectors as Initial States

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It's easy to give a closed-form solution if the initial state is an eigenvector

$$\mathbf{v}_k = A^k \mathbf{v}_0 = \lambda^k \mathbf{v}_0$$

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The Key Point. *This is still true of linear combinations of eigenvectors*

Solutions in terms of eigenvectors

Let's simplify $A^k\mathbf{v}$, given we have eigenvectors $\mathbf{b}_1, \mathbf{b}_2$ for A which span all of $\mathbb{R}^2$:

Eigenvectors and Growth in the Limit

Thm. For a linear dynamical system A with initial state $\mathbf{v}_0$, if $\mathbf{v}_0$ can be written in terms of eigenvectors $\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_k$ of A with eigenvalues $\lambda_1 > \lambda_2 \ldots \geq \lambda_k$ then $\mathbf{v}_k \sim \lambda_1^k c_1 \mathbf{b}_1$ for some constant c_1

In other words, in the long term, the system grows exponentially in the largest eigenvalue

Eigenbases

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Not all matrices have eigenbases

Eigenbases and Growth in the Limit

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for some constant c_1, where where λ_1 is the **largest eigenvalue of** A **and** $\mathbf{b}_1$ **is its eigenvalue**

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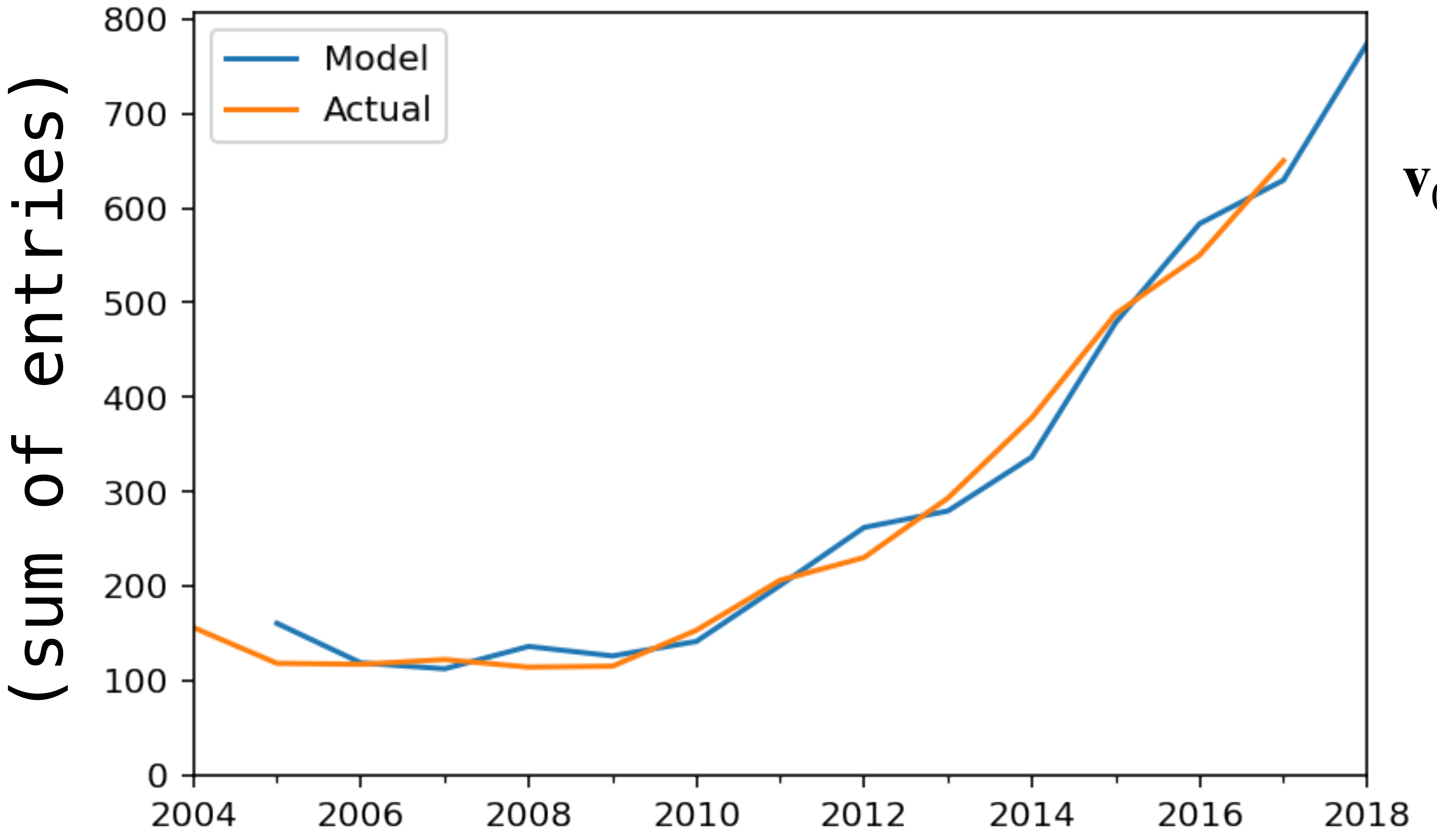
for some constant c_1, where where λ_1 is the **largest eigenvalue of** A **and** $\mathbf{b}_1$ **is its eigenvalue**

The largest eigenvalue describes the long-term exponential behavior of the system

Example: CS Major Growth

see the notes for more details



$$\mathbf{v}_0 = \begin{bmatrix} v_{0,1} \\ v_{0,2} \\ v_{0,3} \\ v_{0,4} \end{bmatrix} = \begin{bmatrix} \text{\# of year 1 students enrolled in 2024} \\ \text{\# of year 2 students enrolled in 2024} \\ \text{\# of year 3 students enrolled in 2024} \\ \text{\# of year 4 students enrolled in 2024} \end{bmatrix}$$

$$\mathbf{v}_k = A^k \mathbf{v}_0$$

(A is determined by least squares)

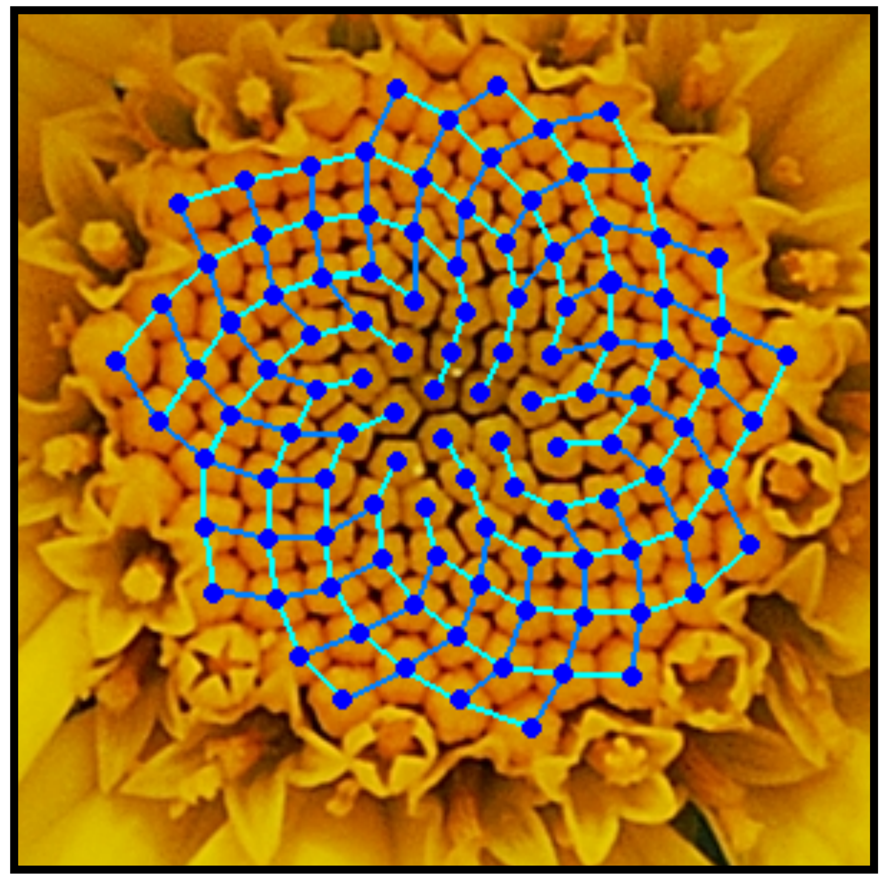
This is clearly exponential. If we want to "extract" the exponent, we need to look at the *largest eigenvalue*

Another Example: Golden Ratio

A Special Linear Dynamical System

$$\mathbf{v}_{k+1} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \mathbf{v}_k \qquad \mathbf{v}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Fibonacci Numbers



$$F_0 = 0$$

$$F_1 = 1$$

$$F_k = F_{k-1} + F_{k-2}$$

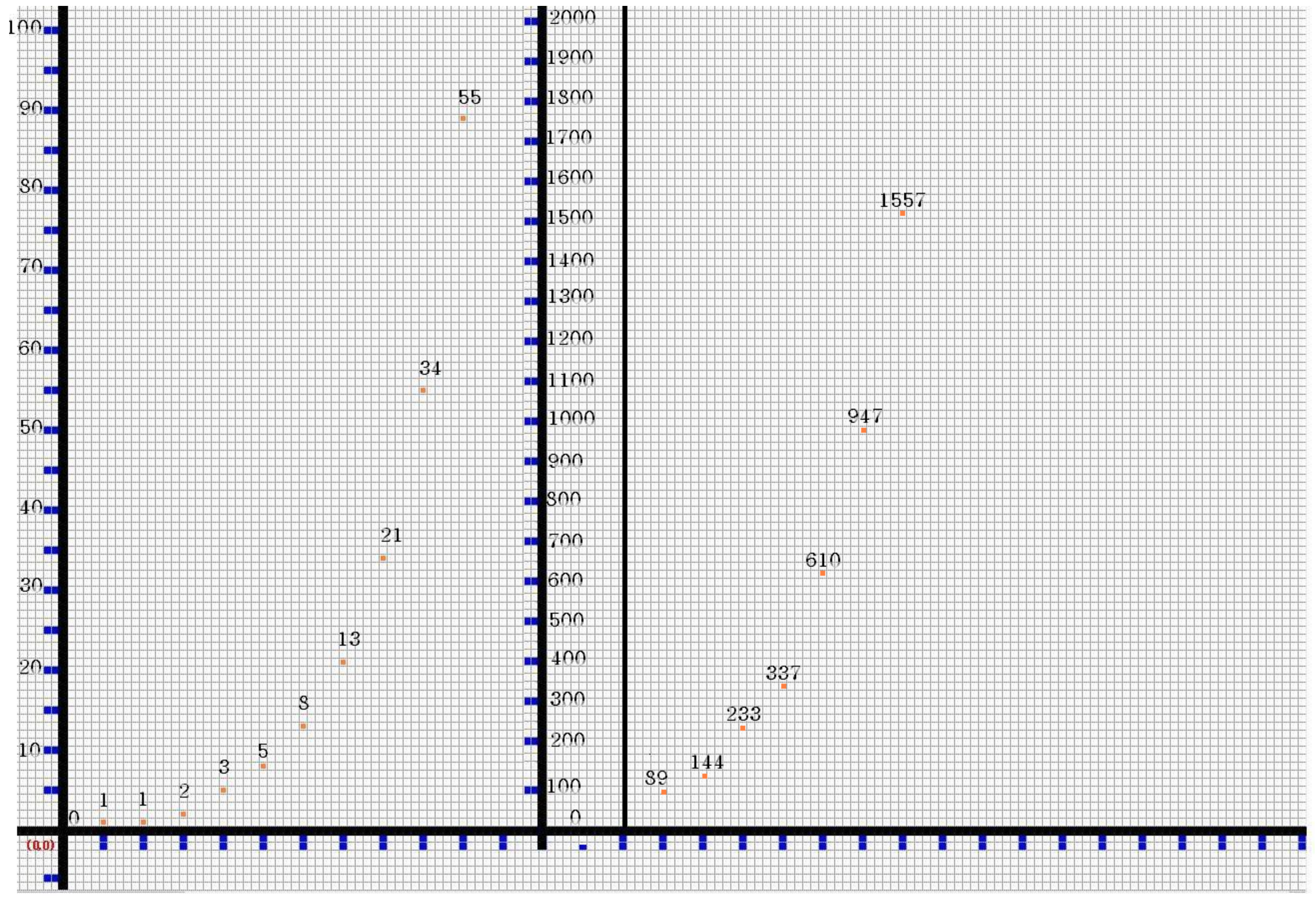
```
define fib(n):
  curr, next ← 0, 1
  repeat n times:
    curr, next ← next, curr + next
  return curr
```

The Fibonacci numbers are defined in terms of a recurrence relation

https://commons.wikimedia.org/wiki/File:FibonacciChamomile.PNG

Golden Ratio

$$\varphi = \frac{1+\sqrt{5}}{2}$$



The "long term behavior" is the Fibonacci sequence is defined by the golden ratio: $F_k \sim \varphi^k$

https://commons.wikimedia.org/wiki/File:Fibonacci_Sequence_Graph.PNG